



A cross-domain knowledge tracing model based on graph optimal transport

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Abstract

Knowledge tracing (KT) is an essential technology in intelligent education that predicts students' performance by analyzing their learning behavior data. Despite significant advances in KT, most existing research focuses on improving performance within a single domain (i.e., a specific academic discipline or school), with limited attention to Cross-Domain Knowledge Tracing (CDKT). CDKT aims to efficiently transfer the capabilities of a KT model from a source domain to a target domain. The main challenge in CDKT is aligning the knowledge state distributions between the two domains to compensate for the lack of interaction data in the target domain. To address this challenge, we propose an Auto-encoder Embedding and Graph Optimal Transport based CDKT model (AEGOT-CDKT). In the initial pre-training phase, we use an auto-encoder to learn the embedding representations of exercises and knowledge concepts in both the source and target domains. Next, we design a cross-domain alignment module that aligns the embedding representations of the two domains using the Graph Optimal Transport (GOT) strategy. This approach effectively supplements students' knowledge state representations acquired from both domains and guides the learning process in the target domain. Additionally, intra-domain alignment is conducted through a graph decoding objective, bringing node features with similar knowledge concepts or exercises closer together. Finally, we train an LSTM model using the aligned source domain embeddings and use the trained LSTM to predict student responses in the target domain. Transfer experiments in different educational contexts demonstrate that our approach successfully accomplishes CDKT task and validates the effectiveness of aligning knowledge state distributions across different domains.

Keywords Knowledge tracing · Graph optimal transport · Cross-domain alignment · Graph attention network

1 Introduction

With the rapid global digitalization and integration of intelligent technologies, smart online education is increasingly pivotal in shaping the education sector. Educational platforms have grown significantly due to digital transformation, providing diverse online learning resources and convenient pathways. While these platforms have revolutionized traditional education

and optimized resource utilization, they still face challenges in meeting personalized learning needs. To address diverse learning demands and vast knowledge, online education platforms must accurately trace and understand students' mastery across various subjects. However, traditional knowledge tracing (KT) methods struggle with data discrepancies between different disciplines and often perform poorly in new subject areas lacking sufficient data. This highlights the necessity for cross-domain knowledge tracing (CDKT). Researchers are now exploring transfer learning to enhance teaching across domains by leveraging knowledge from specific areas.

Transfer learning aims to transfer knowledge from one domain to another to enhance performance in the target domain [1]. Transfer learning can be applied in various domains including computer vision [2], speech recognition [3], equipment fault diagnosis [4–6], and recommendation systems [7]. In the field of KT, Transfer learning techniques are widely applied to address CDKT challenges, aiming to effectively leverage data and knowledge from different domains to enhance model generalization and prediction accuracy. To the best of our knowledge, existing CDKT methods include kernel methods [8], adversarial learning-based methods [9, 10], graph neural network-based methods [11], and domain alignment-based methods [12]. We have decided to combine Graph Neural Networks (GNNs) with domain alignment to achieve CDKT. This is because the field of knowledge tracing involves heterogeneous data such as students, knowledge concepts, and exercises, including complex relationships like students' answer histories, associations between exercises and knowledge concepts, and dependencies among different knowledge concepts, which can naturally be represented using nodes and edges in a graph. We use GNNs to learn features of these nodes. Building upon this, we align feature vectors from two different domains using an explicit distribution distance metric, aiming to minimize domain discrepancies and achieve maximal cross-domain knowledge transfer.

Figure 1 is an example of CDKT, knowledge concept pairs in the mathematical domain (e.g., vector, matrix) and the programming domain (i.e., 1D array, 2D array) are not only semantically similar but also have a progressive relationship in terms of their topological structures. Therefore, they can be matched for knowledge transfer.

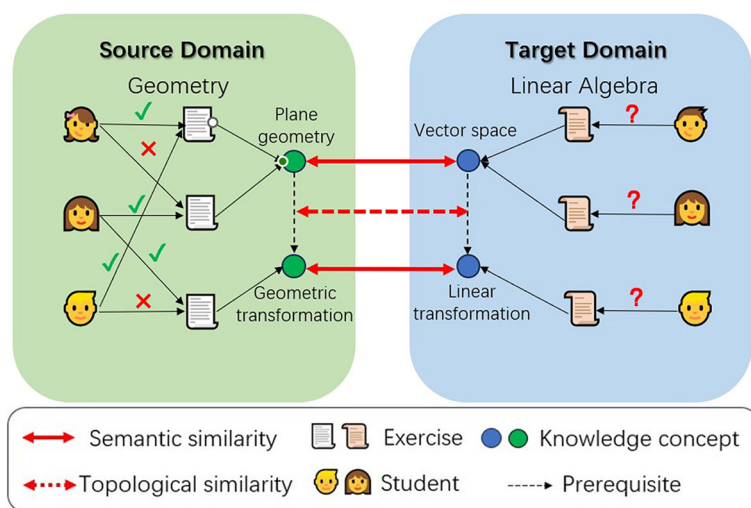


Figure 1 Illustration of aligning nodes and edges from two domains through graph matching

Based on the above considerations, this paper proposes a cross-domain based knowledge tracing, namely AEGOT-CDKT. This approach first utilizes graph auto-encoders to construct knowledge tracing node-attribute graphs in the source and target domains. It captures and quantifies the intricate dependencies and weight allocations among entities within each domain through a graph attention mechanism, thereby generating high-quality embedding vector representations. Subsequently, to facilitate effective knowledge transfer between these two domains, we have devised a cross-domain alignment module that treats the cross-domain alignment issue as a graph matching task. It leverages Graph Optimal Transport (GOT) [13] to establish the correspondence between nodes and edges of the knowledge tracing node-attribute relationship graphs in the source and target domains. Specifically, GOT employs Wasserstein Distance (WD) [14] to measure and match nodes in different domains, ensuring information consistency at the node level. Additionally, introducing Gromov-Wasserstein Distance (GWD) [15] addresses the edge matching challenge, guaranteeing precise alignment of cross-domain data at the edge structure level within the graph. By combining WD and GWD, GOT can achieve maximum matching of corresponding nodes in terms of semantic information and topological structure in the graph, ultimately achieving a more refined cross-domain alignment.

In summary, the contributions of this study are threefold:

- This paper proposes a novel CDKT model based on graph contrasting and transfer learning, which predicts the performance of the next stage of students' learning in the different domain.
- The cross-domain alignment method employed in our model effectively reduces domain distribution discrepancies, facilitating knowledge transfer across different domains.
- This paper reports on the extensive experiments that were conducted in three real datasets. The experiments verify that the proposed model is more effective than the existing KT methods in CDKT task.

The rest of the paper is organized as follows. Section 2 briefly reviews the related work. Section 3 introduce the problem statement. Section 4 describes the framework and the implementation of the proposed approach in detail. Extensive experimental studies to validate the effectiveness of our model and several research questions are addressed in Section 5. Section 6 concludes this work, analyzes the study's limitations and proposes future research directions.

2 Related work

2.1 Traditional knowledge tracing method

Knowledge tracing (KT) relies on real-time interactions between students and exercises to monitor and record their learning progress, thereby evaluating their mastery of knowledge. The academic community has accumulated extensive research findings on KT, which continues to exhibit a consistent momentum of innovation and development. Corbett and Anderson [16] pioneered the Bayesian Knowledge Tracing (BKT) model in 1994, emphasizing the hierarchical structure of knowledge concepts and the structured learning experiences required for knowledge concept mastery [17, 18].

Deep learning technologies, particularly Deep Knowledge Tracing (DKT) introduced by Piech et al. [19] in 2015, have significantly advanced the field. DKT, based on recurrent neural networks, is renowned for its ability to model knowledge states dynamically, offering flexibility and strong predictive performance. Subsequent advancements include Dynamic

Key-Value Memory Networks (DKVMN) by Zhang et al. [20], which utilize memory matrices to store knowledge concepts (Key matrix) and update mastery levels (Value matrix), and MA-LSTM by Li et al. [21], featuring a memory-aware module using temporal attention layers to simulate human brain recall processes in KT. Chen et al. [22] proposed a knowledge tracing model with learning and forgetting behavior (LFBKT), which comprehensively considers factors affecting learning and forgetting. By modeling students' learning and forgetting behaviors, it traces changes in students' knowledge states and predicts their performance. CAKT [23] enhances interpretability by integrating contrastive learning with attention networks. It employs three attention-based encoders to model the dynamic factors of Item Response Theory (IRT) [24] and effectively identify the consistency and separability of knowledge states and learning interactions, leading to improved knowledge state representations.

2.2 Graph-based knowledge tracing method

In educational data science, traditional deep learning methods often convert exercise records into linear knowledge sequences. However, many educational datasets are better represented as non-Euclidean structured graphs [25]. The Graph-based Knowledge Tracing model (GKT), proposed by Nakagawa et al. [26], adopts GNN principles by representing course knowledge concepts as graphs and framing KT problems as time-series node classification tasks within these graphs. This approach has proven effective in KT tasks.

Some KT studies utilize bipartite graphs and employ Graph Convolutional Network (GCN) embedding propagation in graph models with diverse nodes and edges to extract effective exercise representations across heterogeneous knowledge concepts and exercises. For instance, Yang et al. proposed GIKT [27], which utilizes high-order bipartite graphs like "student-exercise" and "student-knowledge concept" to model interactions consistently. Song et al. introduced the "reasoning-generation" model JKT [28], which constructs exercise and knowledge concept graphs and integrates their intrinsic relationships through GCN information propagation. Mao et al. proposed KSGAN [29], which predicts student performance at the exercise level by introducing graph attention layers and optimizing representations. KT-Deeper [30] utilizes dynamic reinforced heterogeneous graph contrastive networks to predict students' cognitive states. It models relationships using temporal heterogeneous graphs and employs reinforcement learning to enhance incomplete graph sequences. Additionally, Song et al. [31] proposed a self-supervised knowledge tracing framework, Bi-CLKT, which leverages bi-graph contrastive learning to capture global and local "Exercise-to-Exercise" (E2E) and "Concept-to-Concept" (C2C) relationships, enhancing the model's performance and interpretability.

2.3 Transfer learning-based knowledge tracing method

In real-world scenarios, KT often encounters challenges such as missing data, cold start, and domain discrepancies, which constrain the effectiveness of traditional KT models. In response to these domain-specific limitations and data sparsity issues, Cheng et al. [8] introduced the three-step AdaptKT strategy. This approach facilitates effective knowledge transfer across domains by leveraging Maximum Mean Discrepancy (MMD) [32] to minimize distribution discrepancies between different domains and fine-tune the output layer.

Huang et al. [9], on the other hand, proposed the STAN framework to address distribution shifts between domains, such as those between different courses, which often degrade model performance when directly applied across domains. They achieve alignment of difficulty

distributions across domains through conditional adversarial learning and difficulty-based sampling methods. However, these methods often overlook the implicit graph topology information present in intermediate layer embeddings and lack detailed pre-training tasks to explicitly model graph topology alignment, relying instead on sequential encoding mechanisms for embeddings. Recent studies [33, 34] have emphasized the critical role of GNNs in explicitly capturing the topological structure of data, exploring both local and global semantics within graphs. This highlights the importance of considering the implicit structure between node embeddings for enhancing KT models.

3 Preliminaries

In this section, we first review the definition of knowledge tracing (KT). Then, we explain what cross-domain knowledge tracing (CDKT) is. We have listed some frequently used symbols in Table 1, and provide a detailed explanation of their roles in this context.

3.1 Knowledge tracing

KT is a data mining task applied in the field of education, which automatically traces the changes in knowledge state based on students' historical learning records. Specifically, the KT task can be defined as follows:

Given a sequence of historical exercise interactions involving specific knowledge concepts by students in an online learning system: $X = (x_1, x_2, \dots, x_t) = \{(e_1, y_1), (e_2, y_2), \dots, (e_t, y_t)\}$, where $x_t = (e_t, y_t)$ typically represents a tuple, e_t is the exercise answered at time t , and $y_t \in \{0, 1\}$ is the answer label indicating whether the exercise was answered correctly or not (1 for correct, 0 for incorrect). KT aims at predict the probability $p_{t+1} = (y_{t+1} = 1 \mid e_{t+1}, X)$ of a student correctly answering the next interaction e_{t+1} .

3.2 Attribute graph of knowledge tracing

To enhance the efficacy of KT predictions, we optimized the reconstruction of model inputs. This involved modeling the multi-dimensional relationships between 'Exercise to Exercise'

Table 1 A summary of notations

Notation	Description
$\mathcal{S}; \mathcal{T}$	Source domain; target domain
$\mathcal{G}_{\mathcal{S}}; \mathcal{G}_{\mathcal{T}}$	Source domain graph; target domain graph
$\mathcal{V}; \mathcal{V} $	A set of exercises; the number of exercises
n	The number of nodes in the graph
$N(i)$	The set of neighbors of node i
$\mathbf{A}$	The structure matrix of graph $\mathcal{G}$
$\hat{\mathbf{A}}$	The reconstruction structure matrix of graph $\mathcal{G}$
x_i	The raw feature vector of node i
$\mathbf{g}_i$	The embedding vector of node i during aggregation
$\mathbf{z}_i$	The final embedding vector of node i after aggregation
$\mathbf{h}_t$	The hidden knowledge state of a student at time step t

(E2E) and ‘Concept to Concept’ (C2C) as separate subgraphs, which were then integrated into a comprehensive knowledge tracing node-attribute graph.

In the Exercise to Exercise (E2E) subgraph $\mathcal{G}_e = (\mathcal{V}_e, \mathcal{E}_e, w_e^{i,j})$, exercises $\mathcal{V}_e$ are represented as nodes, and edges $\mathcal{E}_e$ reflect the relationships between these exercises. By analyzing the correctness of students’ responses to different exercises under the same knowledge concept, we can calculate the influence $w_e^{i,j}$ ($i, j \in \mathbb{N}, i, j \leq m$) between exercises and construct weighted edges accordingly. The weighted influence $w_e^{i,j}$ between two co-occurring exercises is calculated as follows:

$$w_e^{i,j} = \frac{co_correct(\mathcal{V}_e^i, \mathcal{V}_e^j)}{\sum_m co_occur(\mathcal{V}_e^i, \mathcal{V}_e^m)}, \quad (1)$$

where $w_e^{i,j}$ represents the co-correctness rate between exercises $\mathcal{V}_e^i$ and $\mathcal{V}_e^j$. This rate denotes the proportion of students who correctly answered both exercises $\mathcal{V}_e^i$ and $\mathcal{V}_e^j$ among all students who attempted both exercises. It reflects the degree of correlation between the two exercises based on student performance. A higher co-correctness rate suggests that if a student answers one exercise correctly, they are also more likely to answer the other correctly. This metric is useful for identifying exercises that are closely related to each other.

Before constructing the Concept to Concept (C2C) subgraph, we first calculate the average correctness rate of all knowledge concepts in the dataset and compare it with each student’s performance. Students who achieve above-average correctness rates on exercises related to a particular concept are considered to have mastered that concept. Based on this, we then construct the C2C subgraph. Specifically, in the C2C subgraph $\mathcal{G}_c = (\mathcal{V}_c, \mathcal{E}_c, w_c^{i,j})$, knowledge concepts $\mathcal{V}_c$ are represented as nodes, and edges $\mathcal{E}_c$ are determined based on students’ performance on different knowledge concepts. The influence between knowledge concepts $w_c^{i,j}$ is calculated as follows:

$$w_c^{i,j} = \frac{co_mastery(\mathcal{V}_c^i, \mathcal{V}_c^j)}{\sum_m co_occur(\mathcal{V}_c^i, \mathcal{V}_c^m)}, \quad (2)$$

where $co_mastery$ denotes the number of students who demonstrate above-average mastery in exercises related to both knowledge concepts. $w_c^{i,j}$ represents the co-mastery rate between these two knowledge concepts, which is the proportion of students who exhibit above-average mastery in exercises for both concepts. This co-mastery rate reflects the degree of association between the two knowledge concepts, with a higher rate indicating a stronger relationship between the concepts in student learning.

The knowledge tracing node-attribute graph consists of a set of student collections $\mathcal{U}$, exercise set $\mathcal{V}_e$, student and exercise interaction set $\mathcal{E}_u$, and attributes associated with exercises $\mathcal{A}_e$, denoted as $\mathcal{G} = (\mathcal{U}, \mathcal{V}_e, \mathcal{E}_u, \mathcal{A}_e)$, where $\mathcal{U}$ and $\mathcal{V}_e$ serve as vertices, and $\mathcal{E}_u$ serve as edges.

3.3 Cross-domain knowledge tracing

The objective of CDKT is to align the knowledge state distributions of two domains, effectively transferring knowledge from the source domain to the target domain for KT. In the context of transfer learning, we have a more informative source domain $\mathcal{S}$ and a target domain $\mathcal{T}$. Given the historical exercise interaction sequence X of a specific concept for a student, we represent the student’s historical answer records in the source domain $\mathcal{S}$ and in the target domain $\mathcal{T}$ as $X^{\mathcal{S}}$ and $X^{\mathcal{T}}$, respectively. Assuming that we have modeled the multidimensional interaction records between ‘exercise - knowledge concept’ as a graph $\mathcal{G}$, we can describe the

graph generated by the fine-tuned model (target domain model) as the target domain graph $\mathcal{G}_T$, with node embeddings represented as $\{x_1^T, x_2^T, \dots, x_{|\mathcal{V}|}^T\}$; correspondingly, the output generated by the pre-trained model (source domain model) is defined as the source domain graph $\mathcal{G}_S$, with node embeddings shown as $\{x_1^S, x_2^S, \dots, x_{|\mathcal{V}|}^S\}$. To facilitate effective knowledge transfer across domains, a cross-domain alignment module has been designed, treating the cross-domain alignment problem as a graph matching task by matching and adjusting the graph structures of different domains to minimize the differences between the source and target domains in the feature space, making the transfer from $\mathcal{G}_S$ to $\mathcal{G}_T$ smoother and more effective.

4 Methodology

In this section, we will introduce the overall architecture of AEGOT-CDKT and then elaborate on the details of each module.

4.1 Overview of AEGOT-CDKT framework

The overall framework of AEGOT-CDKT is shown in Figure 2. This method mainly includes two stages: pre-training and cross-domain prediction, in which the first stage is the generation of intra-domain exercises and knowledge concept representations, and the second stage is cross-domain prediction. First, in the initial pre-training phase, we use C2C and E2E subgraphs from the source and target domains to learn and optimize interactions between students and knowledge concepts. and is used to train and boost graph autoencoders, resulting in more accurate representations of exercises and knowledge concept embeddings. In the cross-domain prediction stage, we use the LSTM model as the basis to integrate the embeddings obtained in the first stage with the students' learning behavior characteristics as the input of the cross-domain prediction model, so as to construct a comprehensive and complementary representation of the students' knowledge state to predict their performance on the exercises.

We then introduce an innovative alignment module within the LSTM to reduce distribution discrepancies between the source and target domains. This module leverages Graph Optimal Transport to minimize the gap between knowledge state distributions across the two domains, facilitating effective knowledge transfer. This method allows similar learning features from

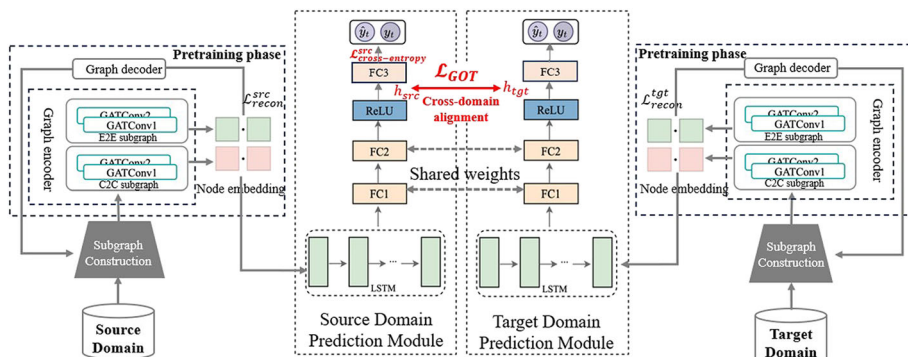


Figure 2 The framework structure of AEGOT-CDKT

the source domain to successfully transfer to the target domain, aiding students in learning the target domain's knowledge.

Subsequently, we integrate learning features from both the source and target domains to construct a comprehensive and complementary representation of students' knowledge states. Finally, we enhance the performance of the cross-domain knowledge tracing model by minimizing cross-entropy loss and the optimal transport distance.

4.2 GAT embedding module

We referred to the pre-training module of AttGAE-KT [35] to obtain initial embeddings of students' knowledge concepts and exercises. Since the GAT embedding process treats both E2E and C2C subgraphs identically, we will simplify the subsequent discussion by not differentiating between them. Specifically, the module first uses a graph attention network (GAT) as an encoder to aggregate the representations of neighbors with different importance in the subgraphs, calculating the attention scores $\alpha_{i,j}$ between center node i and its neighbor j :

$$\alpha_{i,j} = \frac{\exp(\text{LeakyReLU}(a^T [\mathbf{W}\mathbf{g}_i \oplus \mathbf{W}\mathbf{g}_j]))}{\sum_{k \in N(i)} \exp(\text{LeakyReLU}(a^T [\mathbf{W}\mathbf{g}_i \oplus \mathbf{W}\mathbf{g}_k]))}, \quad (3)$$

where $\mathbf{W} \in \mathbb{R}^{F' \times F}$ and $a \in \mathbb{R}^{2F' \times 1}$ represents a linear transformation, $\oplus$ denotes concatenation operation, $\mathbf{g}$ represents the embedding of a node, $N(i)$ represents the set of neighbors of node i , LeakyReLU is an activation function.

Then, the features of neighboring nodes j are weighted and summed using two graph attention convolutional layers and attention scores $\alpha_{i,j}$, resulting in the aggregated embedding $\mathbf{g}_i$ of the neighboring nodes of central node i . For the convolutional layers of GAT at layer $l = 1$, its calculation formula is:

$$\mathbf{g}_i^{(1)} = \sigma \left(\sum_{j \in N(i)} \alpha_{i,j}^{(1)} \mathbf{W}^{(1)} x_j^{(1)} \right), i = 1, 2, \dots, n, \quad (4)$$

where σ represents the *Sigmoid* activation function, x_j is the raw feature vector of node j . For the GATConv layer at layer $l = 2$, its computation process is as follows:

$$\mathbf{g}_i^{(2)} = \sigma \left(\sum_{j \in N(i)} \alpha_{i,j}^{(2)} \mathbf{W}^{(2)} \mathbf{g}_j^{(2)} \right), i = 1, 2, \dots, n, \quad (5)$$

Additionally, to further focus on valuable information from multiple vector subspaces, we introduced the multi-head attention mechanism of GAT to independently perform transformations and average their layer-wise latent representations. Finally, the results after the second layer output are processed by ReLU and used as the final node embeddings.

$$\mathbf{z}_i = \text{ReLU} \left(\frac{1}{H} \sum_{h=1}^H \sum_{j \in N(i)} (\alpha_{i,j}^{(h)} \mathbf{W}^{(2)} \mathbf{g}_j^{(2)}) \right), \quad (6)$$

where H represents the number of parallel attention heads. We used $H = 3$ attention heads to enhance the model's learning capability. $\alpha_{i,j}^{(h)}$ denotes the attention coefficient calculated by the h -th attention mechanism.

In addition to cross-domain alignment, we use a graph decoding strategy to achieve intra-domain alignment of node embeddings. The motivation for intra-domain alignment is to ensure that students within the same domain who exhibit similar levels of knowledge mastery are closer to each other in the embedding space, thus forming more comprehensive and complementary representations. This is crucial for the knowledge tracing task, as students' learning trajectories span multiple domains and require fine-grained modeling of knowledge mastery within each domain. Through this alignment, the consistency of node embeddings within the same domain is enhanced, allowing for more accurate capture of students' performance across different knowledge concepts.

We quantify the similarity in knowledge mastery between students by simplifying second-order relationships to first-order relationships in the source and target domain graphs. This simplification is key in the knowledge tracing task, as it allows us to directly measure the knowledge mastery similarity between two students by embedding their representations in the same space, without considering the complexity of intermediary nodes. Specifically, if two students exhibit similar levels of mastery for many of the same knowledge concepts, their embeddings are adjusted to be more consistent with each other.

To achieve this, we design the following inner product decoder to reconstruct the structure of graph $\mathcal{G}$, without distinguishing between exercises and knowledge concepts for simplicity:

$$\hat{\mathbf{A}}_{ij} = \sigma(\mathbf{z}_i^T \mathbf{z}_j), \quad (7)$$

where σ represents the Sigmoid activation function, $\hat{\mathbf{A}}$ represents the reconstructed structure matrix of graph $\mathcal{G}$, and $\mathbf{z}$ denotes the final node embedding obtained after the second graph attention convolutional layer.

Finally, we consider graph reconstruction as a binary classification task, predicting whether there is a link between two nodes. Formally, we use binary cross-entropy loss as the graph reconstruction loss. Considering that we have both the source domain graph and the target domain graph, the graph decoding loss function for both domains is as follows:

$$\begin{aligned} \mathcal{L}_{dec} &= \mathcal{L}_{src} + \mathcal{L}_{tgt} \\ &= -\frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N (\mathbf{A}_{ij}^S \log(\hat{\mathbf{A}}_{ij}^S) + (1 - \mathbf{A}_{ij}^S) \log(1 - \hat{\mathbf{A}}_{ij}^S)) \\ &\quad - \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N (\mathbf{A}_{ij}^t \log(\hat{\mathbf{A}}_{ij}^t) + (1 - \mathbf{A}_{ij}^t) \log(1 - \hat{\mathbf{A}}_{ij}^t)). \end{aligned} \quad (8)$$

4.3 Prediction module

After obtaining the node embeddings of knowledge concepts and exercises from both the source and target domains, we use them as inputs for the prediction module to determine students' knowledge states in each domain. This prepares the model for cross-domain prediction.

Our prediction module consists of an LSTM layer and two fully connected layers. We chose the LSTM because it effectively captures the temporal features of students' learning processes and can dynamically update their knowledge state based on the sequence in which they answer questions. Specifically, the hidden state of the LSTM is used as the student's knowledge state because it not only incorporates the historical interactions during the learning process but also integrates the most recent learning behaviors, offering a comprehensive and dynamic

representation of the student's knowledge mastery. To achieve this, we first concatenate the knowledge concept embeddings and exercise embeddings obtained through the GAT with the students' external behavior feature vectors. Then, we use the fully connected layer to merge the features and obtain the input $\mathbf{x}$ for the LSTM layer:

$$\begin{aligned}\mathbf{x}_i &= \sigma(\mathbf{W}_1 \cdot (\mathbf{z}_{ci} \oplus \mathbf{z}_{ei} \oplus \mathbf{f}_i)) \\ (\mathbf{h}_t^{\text{lstm}}, \mathbf{c}_t^{\text{lstm}}) &= \text{LSTM}(\mathbf{x}_t, (\mathbf{h}_{t-1}^{\text{lstm}}, \mathbf{c}_{t-1}^{\text{lstm}})),\end{aligned}\quad (9)$$

where σ represents the Sigmoid activation function, $\mathbf{z}_{ei}$ and $\mathbf{z}_{ci}$ represent the embedding vectors of exercise e_i and its corresponding knowledge concept, respectively. $\mathbf{x}_i$ represents the complete embedding of exercise e_i , with each $\mathbf{x}_i$ corresponding to the input at a specific time step t . $\mathbf{x}_t$ represents the input at time step t , $\mathbf{h}_{t-1}^{\text{lstm}}$ and $\mathbf{c}_{t-1}^{\text{lstm}}$ represent the hidden state and cell state of the LSTM unit at time $t-1$. $\mathbf{f}_i = [\mathbf{f}_{i1}, \mathbf{f}_{i2}, \dots, \mathbf{f}_{im}]$ represents the behavioral attributes of a student when completing exercise e_i , where m is the number of features. For the k -th feature in the vector, if it is a categorical feature (exercise type), it is represented in the vector using one-hot encoding as the value $\mathbf{f}_{ik}$; whereas for numerical features (response time), $\mathbf{f}_{ik}$ appears as a scalar value.

Next, the hidden state $\mathbf{h}_{t-1}^{\text{lstm}}$ obtained at time $t-1$ is concatenated with the embedding of the exercise $\mathbf{q}_{next}$ completed at the next time step t to obtain a new feature vector $\tilde{\mathbf{h}}_t$:

$$\begin{aligned}\tilde{\mathbf{h}}_t &= [\mathbf{h}_{t-1}^{\text{lstm}} \oplus \mathbf{q}_{next}], \\ \mathbf{h}_t &= \text{ReLU}(\text{BN}(\mathbf{W}_2 \cdot \tilde{\mathbf{h}}_t) + \mathbf{r}_t),\end{aligned}\quad (10)$$

where $\mathbf{W}_2$ is the weight matrix of the first fully connected layer, $\mathbf{r}_t$ is a portion extracted from the original input $\mathbf{x}_t$ as the residual input, and BN represents batch normalization. Then, the final prediction result is generated through the second fully connected layer:

$$\hat{\mathbf{y}} = \mathbf{W}_3 \mathbf{h}_t, \quad (11)$$

where $\mathbf{W}_3$ is the weight matrix of the second fully connected layer. Finally, we use the cross-entropy loss function to calculate the loss of prediction performance.

$$\mathcal{L}_{\text{pred}} = - \sum_{t=1}^T (\mathbf{y}_t \log(\hat{\mathbf{y}}_t) + (1 - \mathbf{y}_t) \log(1 - \hat{\mathbf{y}}_t)), \quad (12)$$

4.4 Cross-domain alignment

To train an LSTM capable of cross-domain prediction, we begin by acknowledging the need to transfer information from the source domain to the target domain. The model is applied in two scenarios: cross-domain alignment between different stages of the same curriculum and cross-domain alignment of the same curriculum across different schools. In these cases, some knowledge concepts may be shared between the two domains, but their positions within the domain-specific graphs may vary. While the knowledge concepts and exercises differ across domains, resulting in notable variations in the distribution of knowledge states, there are still underlying connections. These include similarities in the hierarchical structure and logical relationships between knowledge concepts, as well as the difficulty and format of exercises. By leveraging these semantic and structural similarities between knowledge concepts and between exercises across domains, we can effectively align knowledge states. Therefore, we use the hidden states of the LSTM as representations of student knowledge states, and calculate the Wasserstein Distance (WD) and Gromov-Wasserstein Distance (GWD) separately

for the knowledge states corresponding to all knowledge concepts in the source and target domains.

Specifically, WD primarily measures the distance between two knowledge states without considering their structural relationships. It focuses on the direct comparison of individual knowledge states, ignoring the broader context of how these knowledge states are arranged or related to each other.

Assuming $\alpha = \sum_{i=1}^n a_i \delta_{h_i^s}$ and $\beta = \sum_{j=1}^m b_j \delta_{h_j^t}$ represent discrete distributions of knowledge states in the source s and target domain t , where $\delta_{h_i^s}$ and $\delta_{h_j^t}$ denote the Dirac delta functions centered at knowledge states h_i^s and h_j^t at their respective time steps i and j . $\Pi(\alpha, \beta)$ denotes the set of all joint distributions $\gamma(h_i^s, h_j^t)$ with marginal distributions $\alpha(h_s)$ and $\beta(h_t)$. $a_i \in \mathbb{R}^{n+}$ and $b_j \in \mathbb{R}^{m+}$ are respectively the weight vectors a_i and b_j , satisfying the normalization condition $\sum_{i=1}^n a_i = \sum_{j=1}^m b_j = 1$. Then, the Wasserstein distance between the two discrete distributions α and β is:

$$\begin{aligned} \mathcal{D}_W[\alpha, \beta] &= \inf_{\gamma \in \Pi[\alpha(h_s), \beta(h_t)]} \mathbb{E}_{(h_s, h_t) \sim \gamma} [c(h_s, h_t)] \\ &= \min_{\mathbf{T} \in \Pi(a, b)} \sum_{i=1}^n \sum_{j=1}^m \mathbf{T}_{ij} \cdot c(h_i^s, h_j^t), \\ c(h_i^s, h_j^t) &= 1 - \frac{(h_i^s)^T \cdot h_j^t}{\|h_i^s\|^2 \cdot \|h_j^t\|^2}, \end{aligned} \quad (13)$$

where $\Pi(a, b) = \{\mathbf{T} \in \mathbb{R}^{n \times m} | \mathbf{T}\mathbf{1}_m = a, \mathbf{T}\mathbf{1}_n = b\}$, $c(h_i^s, h_j^t)$ is the cross-domain cost matrix obtained through cosine distance. $\mathbf{T}$ is the transportation mapping matrix, where $\mathbf{T}_{ij}$ represents the mass transported from h_i^s to h_j^t .

GWD can be applied to align knowledge state spaces by comparing the structural similarity of knowledge representations across different domains. It measures the discrepancies in distances between pairs of knowledge states in the source and target domains. If the distances between corresponding point pairs in the two spaces are nearly identical, it indicates that their overall arrangement and relative positions are also similar. By aligning the distance matrices of these knowledge state spaces, GWD evaluates how well the internal structure of knowledge states in the source domain corresponds to that in the target domain. This is particularly useful for transferring and adapting knowledge tracing models across different educational contexts. Similar to WD, GWD between two discrete distributions α and β is defined as:

$$\begin{aligned} \mathcal{D}_{GW}[\alpha, \beta] &= \inf_{\gamma \in \Pi[\alpha, \beta]} \mathbb{E}_{\substack{(h_s, h_t) \sim \gamma \\ (\hat{h}_s, \hat{h}_t) \sim \gamma}} [L(h_i^s, h_j^t, \hat{h}_i^s, \hat{h}_j^t)] \\ &= \min_{\hat{\mathbf{T}} \in \Pi(a, b)} \sum_{i, \hat{i}, j, \hat{j}} \hat{\mathbf{T}}_{ij} \hat{\mathbf{T}}_{\hat{i}\hat{j}} L(h_i^s, h_j^t, \hat{h}_i^s, \hat{h}_j^t), \\ L(h_i^s, h_j^t, \hat{h}_i^s, \hat{h}_j^t) &= \|c(h_i^s, \hat{h}_i^s) - c(h_j^t, \hat{h}_j^t)\|, \end{aligned} \quad (14)$$

where $L(h_i^s, h_j^t, \hat{h}_i^s, \hat{h}_j^t)$ is a cost function used to evaluate the similarity between two pairs of knowledge states $(h_i^s, \hat{h}_i^s)$ and $(h_j^t, \hat{h}_j^t)$ in their respective domains' relationships and structural characteristics.

In order to enhance domain alignment performance, it is necessary to simultaneously consider the semantic information of knowledge states and their structural information. There-

fore, we refer to the method [13] to combine GWD (denoted by $\mathcal{D}_{GW}$) with WD (denoted by $\mathcal{D}_W$) in order to derive the GOT distance. Within the graph attention convolutional layer, the GOT distance is defined as follows:

$$\begin{aligned} GOT(\mathbf{h}_S, \mathbf{h}_T) &= \xi \cdot \mathcal{D}_{GW} + (1 - \xi) \cdot \mathcal{D}_W \\ &= \sum_{i, \hat{i}, j, \hat{j}} \mathbf{T}_{ij}(\xi \cdot \mathbf{T}_{\hat{i}\hat{j}} L(\mathbf{h}_i^s, \mathbf{h}_j^t, \hat{\mathbf{h}}_i^s, \hat{\mathbf{h}}_j^t) \\ &\quad + (1 - \xi) \cdot c(\mathbf{h}_i^s, \mathbf{h}_j^t)), \end{aligned} \quad (15)$$

where $\mathbf{T}_{ij}$ and $\mathbf{T}_{\hat{i}\hat{j}}$ are transformation matrices for two pairs of knowledge states $(\mathbf{h}_i^s, \mathbf{h}_j^t)$ and $(\hat{\mathbf{h}}_i^s, \hat{\mathbf{h}}_j^t)$, respectively. ξ is a hyperparameter that controls the importance of nodes and edges.

To effectively train a cross-domain model, we combine the LSTM's prediction loss function (12) in the source domain with the domain alignment loss function (15), the final loss function can be defined as:

$$\mathcal{L}_{final} = \gamma \cdot GOT(\mathbf{h}_S, \mathbf{h}_T) + (1 - \gamma) \cdot \mathcal{L}_{src_pred}, \quad (16)$$

where γ is used to control the emphasis on domain alignment versus prediction accuracy during training.

5 Experiments

In this section, we first provide detailed information about the dataset and experiment setup. Then we introduce the extensive experiments which were conducted three domain alignment scenarios and the in-depth analyses. In order to provide a clearer understanding of the model, we address the following key research questions in the subsequent sections:

- RQ1:** Does the model have the capability of cross-domain transfer?
- RQ2:** How do the hyperparameters in the model contribute to the overall performance?
- RQ3:** During the domain adaptation process, what is the distribution of the student's representations in the representation vector space?
- RQ4:** How does the prediction of student's knowledge state change in transfer and non-transfer scenarios?

5.1 Experimental Setup

Datasets To validate the effectiveness of the proposed Cross-Domain Knowledge Tracing method, experiments were conducted on four representative real-world datasets. We summarized the key statistical indicators of the first three datasets in Table 2 for reader reference. Detailed information about the MOOC-Py programming dataset, as outlined in Table 3, was obtained from prior research [36]. To aid in understanding the origins and structures of these datasets, brief overviews of each dataset are provided below:

- **ASSIST2009**¹ This dataset originates from the data resources publicly available on the ASSISTments online tutoring platform during 2009-2010. In this paper, we followed the conventional preprocessing steps adopted in previous works [26, 28, 37], systematically filtering out all records associated with irrelevant attributes containing “noskill” and

¹ <https://sites.google.com/site/assistmentsdata>

Table 2 Dataset statistics

Statistics	ASSIST2009	EdNet-KT1	Matmat
# Students	3,873	5,000	18,847
# Exercises	15,911	10,779	1,150
# Concepts	123	188	433
# Records	432,702	516,105	390,367
# Records per exercise	27.19	47.88	211.35
# Concepts per exercise	1.29	2.21	2.66

answer submission sequences with less than 3 occurrences, as well as exercises labeled as “scaffolding problems”.

- **Matmat** This dataset is one of the datasets collected by the Adaptive Learning Research Team² at the Faculty of Informatics at Masaryk University through its Adaptive Teaching Practice System, which includes a large amount of log information from online math education platforms, recording the interactive behaviors and performances of secondary school students in learning algebra and geometry concepts.
- **EdNet-KT1**³ This dataset is a large-scale dataset constructed from user interaction records on the online educational tutoring platform Santa between 2017 and 2019. The EdNet dataset series is divided into four levels, with each level progressively integrating more learning actions and their subtypes. Due to the abundance of records in KT1, we selected the EdNet-KT1 level. To ensure the validity and feasibility of the experiments, interaction records of 5000 students were randomly selected from this dataset for analysis and modeling.
- **MOOC-Py** This dataset originates from a MOOC online education platform⁴, which offers a course aimed at teaching beginners the basics of Python programming. The dataset contains 9,119 programming instances submitted by beginners while completing six progressively challenging programming exercises. These programming challenges revolve around 10 fundamental Python concepts, including constants (CO), variables (VA), operators (OP), strings (ST), expressions (EX), lists (LI), tuples (TU), dictionaries (DI), conditional statements (CD), and input/output (IO).

Next, we will utilize these preprocessed datasets to evaluate and compare the performance of the modeling techniques proposed in this study.

Data preparation In current research on applying transfer learning to KT, there is a lack of unified standards for dividing data sets into source and target domains. In order to effectively perform CDKT task, following the settings of transfer learning, we adopt a similar approach to the methods in literature [8, 9]. The specific process is as follows:

- First, according to the characteristics of the KT dataset, the data of students, exercises, and knowledge concepts are divided into two independent subsets, representing different source and target domains. Although these two subsets originate from the same original dataset, they exhibit their own characteristic differences in different application scenarios.
- Additionally, to specifically evaluate and validate the effectiveness of knowledge transfer across domains, we considered the influence of contextual factors in student education

² <https://www.fi.muni.cz/>

³ <https://github.com/rriid/ednet>

⁴ <https://www.educoder.net/>

Table 3 Overview of MOOC-Py programming dataset

Challenge	Topic	# Students	# Programs
C1	String concatenation	608	1,038
C2	Modifying a list	553	2,188
C3	Calculating quantities	452	788
C4	Sorting elements	312	1,977
C5	Computing factorials	188	2,236
C6	Modifying a dictionary	72	892
Total	6	2,185	9,119

(such as school affiliation and course topics) on knowledge acquisition and transfer. In the domain of school affiliation, three subsets of data from three different schools were selected from the ASSIST09 dataset as three independent analysis domains, labeled as S1, S2, and S3.

- At the level of course topics, considering the inherent logical progression between topics such as C3 (computing quantities), C5 (computing factorials), C2 (modifying lists), and C6 (modifying dictionaries) on the MOOC-Py platform, which align with the theoretical assumptions of knowledge transfer, these topics were chosen as the objects of investigation to further explore the actual performance of knowledge transfer between course units with clear sequences and dependencies. Basic statistical information is shown in Table 4.

Important to mention, in CDKT task within this paper, the notation ' $\rightarrow$ ' is employed to denote knowledge transfer from the source domain to the target domain.

Comparative models To comprehensively and objectively evaluate the performance of the model proposed in this paper, we designed a series of comparative experiments. We selected both traditional and state-of-the-art deep learning methods as benchmarks. Although these methods are traditionally designed for single domains, they still exhibit some degree of transferability. To apply these models to the Cross-Domain Knowledge Tracing (CDKT) task in this section, each model was first trained on data from the source domain and then directly used to predict outcomes in the target domain. The detailed information of the comparative models is as follows:

- **DKT** [19]. It utilizes the strong sequential modeling capabilities of RNN, especially LSTM, to dynamically trace and predict the mastery of various knowledge concepts by learners at different time steps.

Table 4 Domain transfer data statistics

Domains	# Students	# Exercises	# Concepts
S1	231	11,692	101
S2	322	6079	54
S3	336	7187	85
C2	553	2188	6
C6	72	892	6
C3	452	788	5
C5	188	2236	7
Total	6	2,185	9,119

Table 5 Random partitioning of student data

Student set	Shared subset	Source subset	Target subset
U	U_{shared}	$U_{\text{shared}} \cup U_{\text{d1}}$	$U_{\text{shared}} \cup U_{\text{d2}}$
Control distribution shift		Control similarity and dissimilarity	

- **DKVMN** [20]. It utilizes two memory matrices to store learners' short-term and long-term memory states for each knowledge concept, and combines differential attention mechanisms to update these memory units.
- **GKT** [26]. Its core lies in simulating the inherent connections between various knowledge concepts through constructing a knowledge concept relationship graph, and innovatively transforming traditional KT problems into a task of node-level classification on a time series.
- **JKT** [28]. Proposing the "Inference-Generation" model, leveraging Graph Convolutional Networks to propagate embedded information and integrate the inherent connections between questions and knowledge concepts.
- **KSGAN** [29]. It incorporates graph attention layers and representation optimization to predict students' performance at the exercise level.
- **AttGAE-KT** [35]. It utilizes attention graph auto-encoders to model exercises and knowledge concepts in KT tasks, delving into the multi-dimensional subgraph relationships between exercises, knowledge concepts, and their subgraphs.

Experimental details Our model is implemented using the PyG⁵ framework, and experiments were conducted on a Linux server equipped with two NVIDIA A10 GPUs. During the training process, we utilized Adam as the optimizer with a learning rate set to 0.0001, a batch size of 256, and 200 epochs. The model comprises 2 graph attention convolutional layers with 3 attention heads. The embedding dimensions for exercises and knowledge concepts are both 64, while the hidden layer dimension is 128.

We adopted the performance evaluation criteria commonly used by most KT institutes, selecting Area Under the Curve (AUC) as the metrics for assessing model performance. AUC measures the area under the ROC curve, plotting true positive rate against false positive rate. It provides an aggregate performance measure across classification thresholds. The improvement in AUC values reflects the enhancement of the model's predictive capability.

5.2 Domain transfer experiments (RQ1)

Transfer with random partitioning Due to the general lack of specific background information for students in the dataset, we adopted the method of randomly partitioning data to construct the transfer learning scenario. Specifically, we labeled all student data as set U , and selected 20% of the student data as the shared student set, denoted as U_{shared} . The remaining student data was divided into two disjoint subsets, denoted as U_{d1} , U_{d2} , with $U_{\text{d1}} \cap U_{\text{d2}} = \emptyset$. Furthermore, by combining the shared student set with these two subsets, we obtained $U_{\text{shared}} \cup U_{\text{d1}}$ and $U_{\text{shared}} \cup U_{\text{d2}}$, which respectively formed the student groups in the source domain and the target domain. This partition strategy aims to simulate cross-domain transfer learning scenarios in the real world, where student groups have both similarities and differences in the source and target domains. Table 5 documented the random partitioning

⁵ <https://pytorch-geometric.readthedocs.io/>

Table 6 Comparison of KT models' performance in cross-domain transfer among students under different distribution transfer forms (AUC)

Model	ASSIST09		Matmat		Ednet-KT1	
	Src→Tgt	Tgt→Src	Src→Tgt	Tgt→Src	Src→Tgt	Tgt→Src
DKT	0.6582	0.6445	0.6727	0.6623	0.6505	0.6686
DKVMN	0.7590	0.7419	0.7381	0.7204	0.7314	0.7221
GKT	0.7305	0.7417	0.7214	0.7257	0.7369	0.7398
JKT	0.7981	0.7787	0.8188	0.7898	0.7252	0.7244
KSGAN	0.7735	0.7706	0.8488	0.8223	0.7687	0.7669
AttGAE-KT	0.8135	0.7940	0.8612	0.8331	0.8243	0.8116
AEGOT-CDKT	0.8201	0.8227	0.8832	0.8628	0.8256	0.8173

of student data. Similarly, we also partition exercises and knowledge concepts according to this strategy.

Three types of cross-domain knowledge tracing tasks were constructed for different transitions of students, exercises, and knowledge concepts. Tables 6, 7, and 8 show the knowledge tracing results of AEGOT-CDKT and all comparative models. In the transfer learning experiments of this section, bidirectional transfer evaluations were conducted on the ASSIST09, Matmat, and EdNet-KT1 datasets, specifically transferring from the source domain to the target domain (Src → Tgt) and from the target domain to the source domain (Tgt → Src). In particular, for different transfer task settings:

- From Table 6, it can be observed that the AEGOT-CDKT model performs best in CDKT task among randomly grouped students, especially showing significant leadership on the Matmat and EdNet-KT1 datasets.
- Table 7 highlights the strong performance of the AEGOT-CDKT model in the transfer task of “random partition of exercises”, especially in predicting target tasks on the ASSIST09 and Matmat datasets. Its AUC values are notably higher compared to other single-domain models.
- Table 8 reveals that single-domain models suffer a substantial decline in cross-domain performance when tackling more intricate knowledge concept transfer issues, owing to their limited transferability. In contrast, the AEGOT-CDKT model retains its leadership position, highlighting its capacity to harness universal information embedded within

Table 7 Comparison of KT models' performance in cross-domain transfer among exercises under different distribution transfer forms (AUC)

Model	ASSIST09		Matmat		EdNet-KT1	
	Src→Tgt	Tgt→Src	Src→Tgt	Tgt→Src	Src→Tgt	Tgt→Src
DKT	0.6528	0.6416	0.6892	0.6762	0.6569	0.6600
DKVMN	0.7223	0.7090	0.7117	0.7236	0.7167	0.7119
GKT	0.7230	0.7306	0.7463	0.7345	0.7324	0.7576
JKT	0.7655	0.7590	0.8011	0.7902	0.7691	0.7684
KSGAN	0.7812	0.7783	0.8225	0.8060	0.7779	0.7756
AttGAE-KT	0.8128	0.7992	0.8373	0.8320	0.8202	0.8124
AEGOT-CDKT	0.8316	0.8362	0.8815	0.8621	0.8362	0.8359

Table 8 Comparison of KT models' performance in cross-domain transfer among knowledge concepts under different distribution transfer forms (AUC)

Model	ASSIST09		Matmat		EdNet-KT1	
	Src→Tgt	Tgt→Src	Src→Tgt	Tgt→Src	Src→Tgt	Tgt→Src
DKT	0.6671	0.6462	0.6847	0.6781	0.6802	0.6941
DKVMN	0.6914	0.6907	0.6961	0.7023	0.7184	0.7258
GKT	0.6882	0.7395	0.7344	0.7276	0.7339	0.7264
JKT	0.7712	0.7654	0.7992	0.7853	0.7609	0.7613
KSGAN	0.7921	0.7505	0.8256	0.8027	0.7752	0.7749
AttGAE-KT	0.7724	0.7685	0.8241	0.8223	0.8062	0.7873
AEGOT-CDKT	0.8123	0.8114	0.8789	0.8556	0.8327	0.8245

source domain knowledge concepts to offset the deficiencies in target domain information.

The results indicate that whether it's student groups, exercises, or knowledge concepts being transferred, the AEGOT-CDKT model is able to more effectively capture and utilize source domain information, enhancing learning outcomes in the target domain. This is attributed to the model's optimization in feature representation, attention mechanisms, and cross-domain alignment, enabling it to have better adaptability and generalization in the field of knowledge tracing. In other words, the model can better adapt to new environments and scenarios during the transfer learning process.

Transfer based on education context The experiment investigated the effect of model transfer in educational environments, particularly in terms of the ability to predict knowledge states across different schools and course topics. Tables 9 and 10 show the detailed results of the transfer experiments conducted in different school domains and course topic domains. The results of the transfer experiments in the school domain indicate that when the model is applied to predict the knowledge mastery of students across different schools, the AEGOT-CDKT model performs excellently in all transfer scenarios. In particular, in the scenarios of transferring from S1 to S3 and from S3 to S1, the transfer accuracy of the AEGOT-CDKT model is significantly higher than that of other reference models.

In Table 10, we analyzed the transfer effects between different course topics. These course topics, such as C3 (computing quantities), C5 (computing factorials), C2 (modifying lists), and C6 (modifying dictionaries), exhibit logical progression in content. The AEGOT-CDKT model demonstrates commendable performance in transfer learning between these topics. For instance, in the transfer from C2 to C6, i.e., from list modification knowledge concepts to dictionary modification knowledge concepts, AEGOT-CDKT effectively utilizes the mastered knowledge of list operations to facilitate learning of dictionary operations. Similarly, in

Table 9 The experimental results of cross-domain transfer among different schools

Model	S1→S2	S2→S1	S1→S3	S3→S1	S2→S3	S3→S2
DKT	0.6650	0.6518	0.6706	0.6681	0.6231	0.6218
JKT	0.7183	0.7009	0.7369	0.7018	0.7273	0.7241
KSGAN	0.7061	0.7036	0.7639	0.7341	0.6918	0.6892
AttGAE-KT	0.7321	0.7146	0.7751	0.7498	0.7419	0.7295
AEGOT-CDKT	0.7381	0.7404	0.7792	0.7795	0.7431	0.7413

Table 10 The experimental results of cross-domain transfer among different schools

Model	C2→C6	C6→C2	C3→C5	C5→C3
DKT	0.6335	0.6374	0.6282	0.6320
JKT	0.6898	0.6824	0.7086	0.6891
KSGAN	0.6692	0.6710	0.7295	0.7059
AttGAE-KT	0.7063	0.7052	0.7431	0.7194
AEGOT-CDKT	0.7110	0.7129	0.7474	0.7403

the transfer from C3 to C5, AEGOT-CDKT leverages the knowledge concepts of computing quantities to reinforce the learning of computing factorials.

Thus, this further reflects that our proposed cross-domain alignment module can effectively enhance the model's generalization ability and adaptability across domains.

5.3 Sensitivity analysis (RQ2)

In CDKT task, the selection of hyperparameters is paramount as they directly influence the model's performance and its ability to generalize on the target domain. Therefore, we assess their impact on the model's predictive performance by adjusting the values of γ and ξ . γ is crucial for balancing the model's predictive accuracy on the source domain with the effectiveness of the GOT in transferring knowledge to the target domain. Meanwhile, ξ plays a critical role in adjusting the relative importance of two distinct distance metrics within the GOT framework. To systematically evaluate their effects, we fix one hyperparameter at a time and systematically vary the other across values of 0, 0.25, 0.5, 0.75, and 1. This rigorous evaluation is conducted across three diverse datasets using CDKT task, which partitions knowledge based on concepts.

From Figure 3, it can be observed that when $\gamma = 0.25$, the AUC score is maximized. This signifies that $\gamma = 0.25$ achieves the best cross-domain performance by effectively aligning the knowledge state distributions between the source and target domains, while also ensuring reasonable convergence of training loss in the source domain. An increase in γ diminishes the weight assigned to the GOT, thereby compromising the model's transferability. Similarly, when $\xi = 0.25$, the model exhibits optimal performance, indicating that in the field of knowledge tracing, the role of Wasserstein Distance (WD) is more important than that of Gromov-Wasserstein Distance (GWD). Specifically, WD focuses on the absolute distances between knowledge states, which helps ensure semantic matching and alignment of knowledge states. This is crucial for enhancing cross-domain adaptability as it directly aligns the actual content of knowledge. In contrast, GWD focuses on the relative structure between knowledge states, primarily comparing the topological structures of the spaces. Although GWD is useful for capturing the structural information of knowledge states, its impact on overall transfer performance in knowledge tracing tasks is relatively smaller. This suggests that during domain transfer, accurate alignment of semantic information (via WD) is more critical than the alignment of structural information (via GWD). Therefore, WD more effectively captures semantic matching between knowledge states, significantly improving the model's cross-domain adaptability.

5.4 Distribution Alignment Visualization (RQ3)

We utilized the T-SNE algorithm [38] to conduct dimensionality reduction on the features obtained from different disciplinary domains and course topics for visualization in a two-

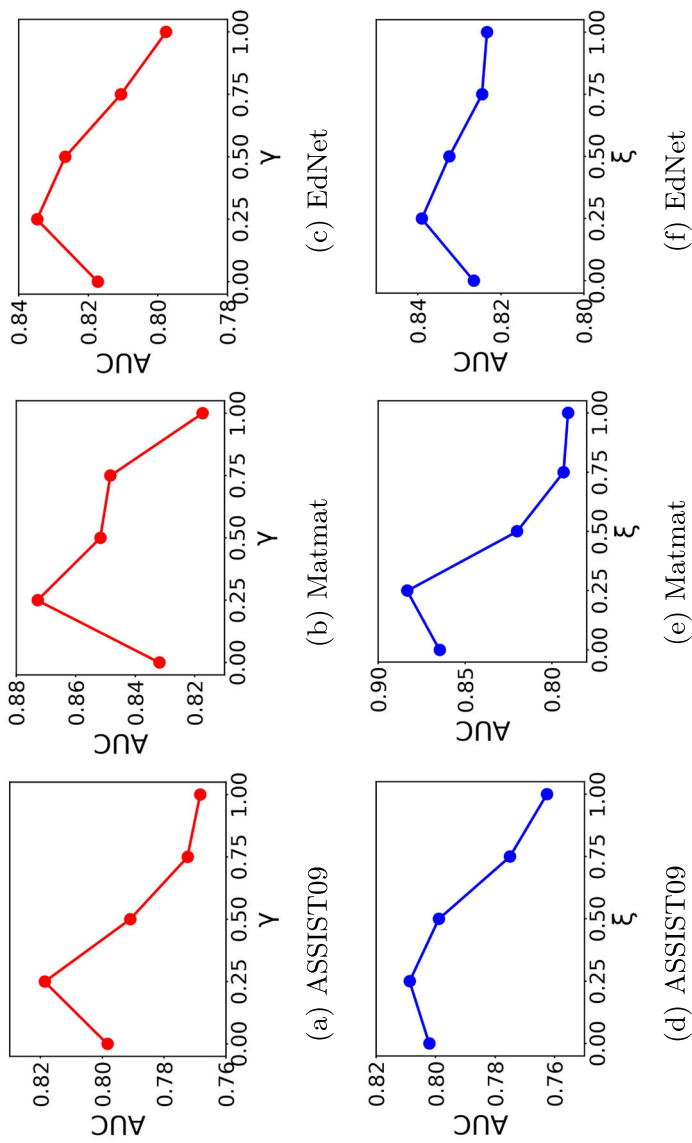


Figure 3 The impact of hyperparameters γ and ξ on AUC scores

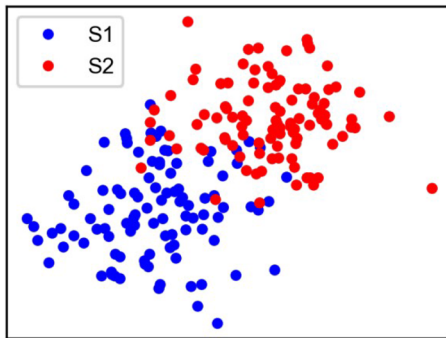
dimensional space. Combining with the visualization results in Figure 4, it can be observed that the distribution of knowledge states between different domains exhibits a certain degree of dispersion in the two-dimensional space, indicating significant differences. Further analysis reveals that, at the course topic level, the distribution distance of students' knowledge states between C3 (calculating quantities) and C5 (computing factorials) is smaller than that between C1 (string concatenation) and C5. This is because C3 and C5 have logical coherence in their content, whereas C1 and C5 differ significantly in both content and skill requirements, resulting in weaker correlation. At the school level, the significant differences in students' learning abilities and habits across different schools lead to larger distribution distances in knowledge states when comparing across schools. After training with the cross-domain alignment module and intra-domain alignment mechanism, the knowledge states of different schools or course topics become more concentrated and similar, with some overlap. This result demonstrates the effectiveness of our proposed method in reducing the discrepancies in knowledge state distribution between different disciplinary domains and course topics, thereby facilitating cross-domain knowledge transfer.

5.5 Stability Analysis (RQ4)

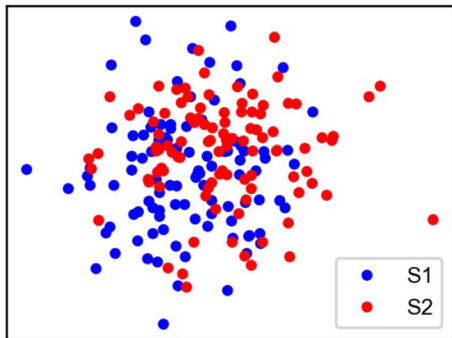
In this section, we analyzed the knowledge state changes of students under the transfer and non-transfer scenarios using the AEGOT-CDKT model. Figure 5 illustrates the knowledge state changes of knowledge concepts c32 (positive decimal ordering) and c33 (fraction ordering) when transferring between schools S1 and S2. The horizontal axis of the knowledge state change graph represents a student's sequence of interactions, where each time step consists of (concept, answer), indicating the student's answering situation at a certain time step. The vertical axis lists 2 different knowledge concepts. The depth of color in the graph corresponds to the model's forecasted probability of a student providing a correct answer. Darker shades signify a greater probability according to the model that the knowledge concept has been correctly understood.

We observed that in the non-transfer scenario in Figure 5, the model's answer predictions exhibited some fluctuations, indicating that the model only utilized fewer data in the target domain S2, neglecting the nonlinear relationships between students, knowledge concepts, and exercises, resulting in a lack of deeper semantic understanding of the students' knowledge state. After domain transfer, the student's knowledge state showed a stable trend without any abnormal fluctuations. After answering correctly (or incorrectly) in a concept's exercise, the corresponding concept mastery level gradually increased (or decreased). It can be observed that in the first three exercises of c32 (positive decimal ordering), the student only answered 1 correctly, indicating a relatively low mastery level of "positive decimal ordering" initially; however, in the later 5 exercises of "positive decimal ordering", the student answered 4 correctly, leading the model to believe that he gradually mastered "positive decimal ordering". Similarly, although the student's overall accuracy in c33 (fraction ordering) is relatively high, he answered incorrectly in 2 out of the last 4 exercises of "fraction ordering", causing the model to determine that the student only partially mastered "fraction ordering" in the end.

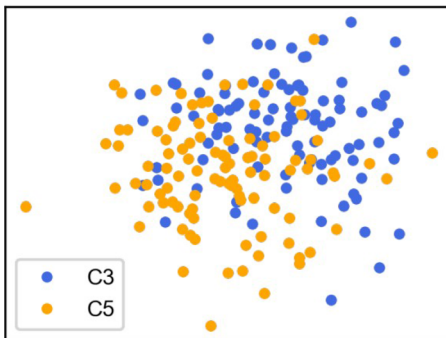
Overall, the model's judgments align with our understanding. This phenomenon suggests that the model can improve its knowledge state learning ability in the target domain through the representation of knowledge states in the source domain after transfer. Thus, our approach indeed facilitates knowledge state sharing and helps the model better predict in new domains.



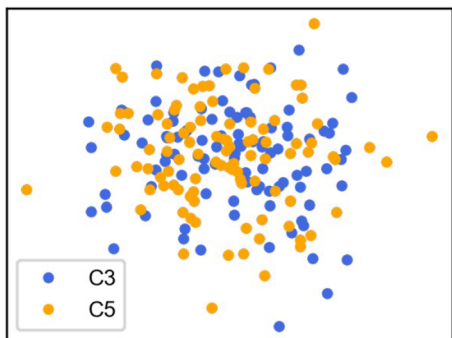
(a) Feature Distribution Before Alignment Among Schools



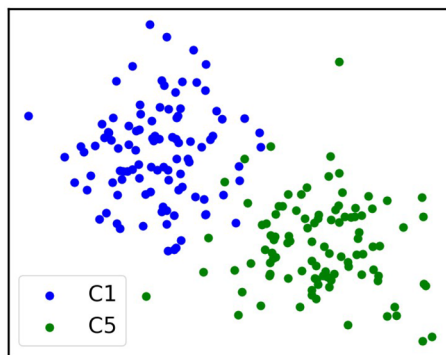
(b) Feature Distribution After Alignment Among Schools



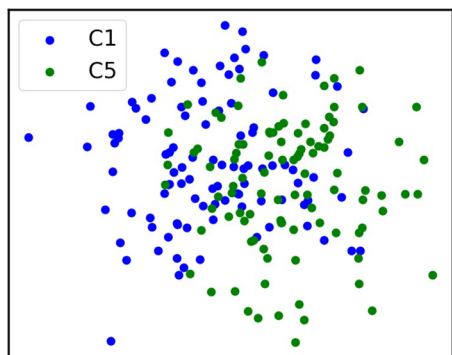
(c) Feature Distribution Before Alignment Among Similar Topics



(d) Feature Distribution After Alignment Among Similar Topics



(e) Feature Distribution Before Alignment Among Weakly Related Topics



(f) Feature Distribution After Alignment Among Weakly Related Topics

Figure 4 The visualization of the alignment results of knowledge state distributions between the source and target domains

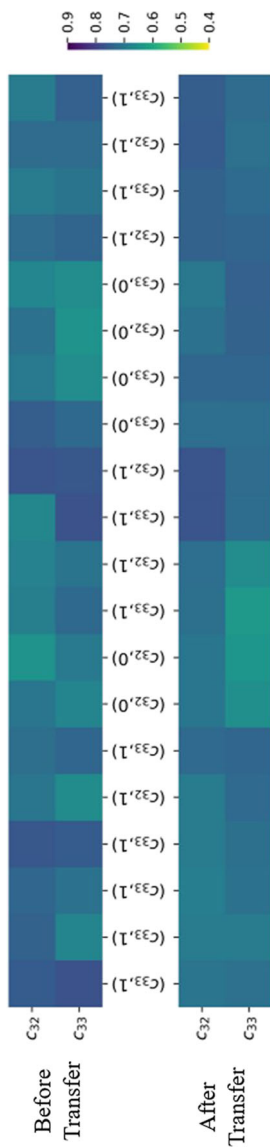


Figure 5 The variation of a certain student's knowledge state in the transfer and non-transfer scenario

6 Conclusions

This paper presents a cross-domain knowledge tracing model based on transfer learning called AEGOT-CDKT, aiming to bridge the knowledge gap between different domains while optimizing and enhancing the performance of knowledge tracing in new domains. The core innovation of AEGOT-CDKT lies in introducing an alignment module capable of handling and reducing the distributional differences of learners' knowledge states across different domains. Moreover, it applies Graph Optimal Transport algorithm to effectively align the graph structures of different domains, thus enhancing the complementarity of student knowledge state representations and providing learners with more precise learning guidance.

Finally, experimental studies and visual analyses are conducted to validate the models excellent performance in various target domains, further demonstrating the crucial role of transfer learning in improving the generalization and adaptability of knowledge tracing models.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethical Approval This declaration is not applicable

Competing of Interest The authors declare no competing interests.

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